



Facebook's modular phone

A patent application details that Facebook's secretive division is reportedly developing a modular smartphone. <http://dgit.in/modularFB>



Whiskers for robots

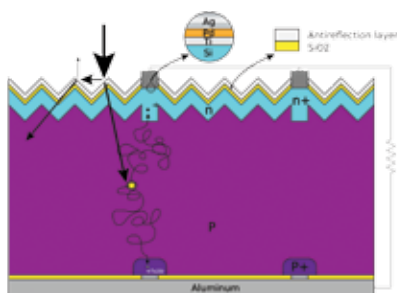
A mechanical engineer from Northwestern University is working on whiskers for robots for a powerful new sense. <http://dgit.in/whiskerBot>



Parabolic troughs are used to harness the thermal energy of the sun and use it for various purposes - ranging from cooking to power generation. Check out <http://dgit.in/CSPtrough> to understand how they work.

due to his discovery of the photovoltaic effect in 1839, which he explained as “the production of an electric current when two plates of platinum or gold immersed in an acid, neutral, or alkaline solution are exposed in an uneven way to solar radiation.”

It was the discovery of photoelectric effect in selenium by William Grylls Adams and his student Richard Evans Day that kickstarted the modern concept of a solar cell. Within the next decade, in 1883, Charles Fritts actually made the first solar cells from selenium wafers. This led to the credit for the invention of the solar cell majorly going to Fritts over time. Modern solar cells, however, are made of silicon, not selenium. The silicon solar cell was developed by Daryl Chapin, Calvin Fuller, and Gerald Pearson at Bell Labs in 1954. This was the first time photovoltaic cells were able to produce enough electricity to reasonably power small electronic devices. The New York Times was perhaps the most accurate in summing up the public sentiment around this discovery by calling it “the beginning of a new era, leading eventually to the realisation of harnessing the almost limitless energy of the sun for the uses of civilisation.”



A solar cell's structure.

But a major restriction in its initial days was the cost. In 1956, the commercially available solar cells would cost around \$300 per watt. In spite of this, certain novelty items like toys and radios were beginning to include solar cells into their builds.

In fact, another area where solar saw rising usage was satellites. The Vanguard 1, launched in 1958, was the first satellite to use solar power – it contained a small one watt array to power its radios. Due to the ample availability of direct sunlight in space, this implementation picked up and later that year, the Vanguard 2, The Explorer 3 and Sputnik-3, all went to space with solar arrays on board. By 1964, NASA had launched Nimbus, a satellite powered entirely by a 470-watt solar array, into a sun-synchronous orbit to make most of the solar power available in space. Encouraged by the success of these launches, NASA launched an Orbiting Astronomical Observatory, powered by a one-kilowatt array, in 1966.

If you think that Tesla (The Elon Musk-led automobile, power and solar company) was the first to implement the concept of a solar roof, you would be wrong. Although they are one of the companies that have made the idea commercially viable as a product, the University of Delaware had incorporated this idea into the design of their “Solar One” solar building in 1973. The building ran on a hybrid of solar-ther-

mal and solar-photoelectric power. Even with these developments, solar energy was little more than an experiment in the labs that was reserved for advanced implementations like the satellite. It was the oil and energy crisis in the 1970s that woke the world up to the need for an alternate, renewable energy source and solar energy was the one readily available. Gradually, focus shifted towards increasing the efficiency of generating solar energy.

It wasn't until 1985 that an efficiency of 20% was achieved. M. A. Green, A. W. Blakers, S. R. Wenham, S. Narayanan and M. R. Willison formed the team that demonstrated the achievement of 20% efficiency under terrestrial conditions.

The next time this was improved upon was in 1999 by the National Renewable Energy Laboratory in collaboration with SpectroLab Inc. by achieving an efficiency of 33.3%, which was again surpassed in 2016. Researchers from the University of South Wales once again improved upon solar cell efficiency by achieving a conversion rate of 34.5%.

DID YOU KNOW

The Solar Orbiter 2 is the most powerful solar powered aircraft out there - it has even circumnavigated the globe and registered the first zero-emissions flight to do so!

TODAY

Today, the world is interested in solar power like never before. Companies like SolarCity (now a part of Tesla) have brought solar power into the hands of people around the world. Offering products like the solar roof, electric vehicle chargers, battery packs and commercial solar installation services, companies like these have played a huge role into bringing the cost of solar power down to the nominal \$0.50 (anywhere between Rs. 30 - 50 per watt in India, depending on the size and efficiency of your solar setup) that it costs today.

We might be looking at a time where solar power supersedes other power sources, especially non-renewable energy. Going by the current support behind it, both from the government and the industry, that time might be sooner than later. 